

RESEARCH ARTICLE OPEN ACCESS

Reliability Analysis of Electricity Grid Integrated With PV System and BESS at Distribution Level: A Case Study of Ntongwe and Gatumba Feeders at Rwanda's Electricity Network

Obed Nkuriyingoma^{1,2}  | Engin Özdemir¹ 

¹Department of Energy Systems Engineering, Kocaeli University, Kocaeli, Türkiye | ²Department of Mechanical and Energy Engineering, University of Rwanda, Kigali, Rwanda

Correspondence: Obed Nkuriyingoma (nkuriyobas@gmail.com)

Received: 20 August 2024 | **Revised:** 12 June 2025 | **Accepted:** 10 July 2025

Keywords: battery energy storage system (BESS) | DigSILENT PowerFactory | grid reliability | solar PV system

ABSTRACT

The escalating energy demand in today's world underscores the need for a consistent and continuous power supply. There is a push towards promoting renewable energy sources to address this demand because of their positive impact on the environment and their widespread accessibility. Particularly in Rwanda, for the radial distribution network, the reliability improvement in electricity supply requires rectifications of power system infrastructure, including transmission, distribution system, and an increase in energy generation capacity. To evaluate the influence of renewable energy sources (RES) on the reliability of Rwanda's power grid, Solar Photovoltaic (PV) systems combined with Battery Energy Storage Systems (BESS) are introduced into the Gatumba and Ntongwe feeders. For the reliability enhancement of the grid to the feeders where the solar PV system and BESS are connected, protection switches are configured so that the PV systems and BESS supply the load demand in the absence of the grid, thereby increasing the generation capacity and reducing interruption frequency and duration. This study assesses how the integration of solar PV plants with BESS can improve the reliability of Rwanda's electricity grid, specifically at the Gatumba and Ntongwe feeders. Simulation results from DigSILENT and PV*SOL simulation tools showed that by integrating solar PV systems with BESS to Gatumba and Ntongwe feeders, the system reliability ameliorates by 71.6% and 95.5% for system average interruption frequency and duration, respectively.

1 | Introduction

The significance and requirement of energy in the contemporary world are growing steadily. The rising demand for energy underscores the importance of a reliable and consistent supply. Nonetheless, the escalating energy use from traditional sources raises environmental and global warming concerns. As a result, eco-friendly alternative energy sources are being promoted and adopted to mitigate the negative effects of conventional energy

sources and tackle the climate crisis. This shift is driven by the associated benefits of renewable sources for the environment, energy accessibility, and sustainability [1]. Solar photovoltaic (PV) panels are increasingly being used for electricity generation due to the widespread availability of solar energy, the simplicity and environmental benefits of PV systems, and the strong government support. The recent decrease in the cost of energy production has further accelerated the adoption of solar PV systems, making them a cost-effective option for electricity

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Engineering Reports* published by John Wiley & Sons Ltd.

generation in applications ranging from small to large scale [2–4]. Battery energy storage systems (BESS) are utilized to ensure a consistent energy supply and mitigate the intermittent nature of solar energy. During periods of low energy demand, batteries store surplus energy generated by PV panels. When the sun's energy is insufficient to meet the power demand, such as during nighttime, the stored energy is utilized. Integrating batteries enhances the reliability and self-sufficiency of the solar PV system [5–8]. Incorporating PV systems into the grid presents several difficulties, such as predicting output power, maintaining voltage stability, responding to frequency changes, providing reactive power support, and ensuring power quality. Overcoming these challenges requires developing systematic solutions such as grid codes and policies, advanced control strategies, and energy storage systems [9, 10]. On the other hand, grid-connected solar PV systems play a key role in enhancing the electrical grid's reliability by improving overall system performance. This improvement is achieved through optimizing system size and strategic placement of solar PV installations [11, 12]. This study introduces the other applications of PV systems not only for generating electrical power but also for improving the reliability of the electricity grid in developing countries.

This research assesses how the integration of solar PV plants with storage systems can improve the reliability of Rwanda's electricity grid, specifically at the distribution level of the Gatumba and Ntongwe feeders. It also examines the effects of integrating Solar PV power plants with BESS systems on Rwanda's electricity grid from various perspectives. Gatumba and Ntongwe feeders are modeled in DigSILENT PowerFactory, and the Solar PV systems and BESS power flow profiles are simulated using PV*SOL software. The paper is organized as follows: Subsequent to the introduction, Section 2 offers background information on the Rwanda electricity grid and examines relevant studies. Section 3 outlines the methodology for system configuration, data collection, and simulations. Section 4 discusses simulation results for various scenarios, and Section 5 offers a conclusion.

2 | Background and Literature Review

2.1 | Rwanda Energy Overview

Rwanda, situated in the eastern region of Africa, covers a total land area of 26 338 Km² and has a population exceeding 13 million. The Rwandan government aims to transform Rwanda from a developing country to a middle-income country by 2035 through the National Strategy for Transformation (NST1). Rwanda's government has committed to achieving universal access to electricity by 2024. This ambitious objective aligns with the 7th Sustainable Development Goal (SDG), which seeks to guarantee affordable, dependable, and clean energy access. This initiative reflects Rwanda's strong commitment to providing its population with clean and affordable energy, thereby contributing to economic development and improved quality of life for its citizens. This target comes after the country recorded an 18% access rate in 2014, which increased to 65.7% by June 2023 [13].

2.1.1 | Electric Power Generation

Rwanda's electricity is generated from various sources, including hydroelectric power, methane gas, peat, thermal, and solar energy. The total installed capacity in Rwanda stands at 353.4 MW. The estimated access rate to electricity in Rwanda is 71.9%, with 53.6% of the population having access to on-grid electricity and 18.3% having access to off-grid sources. The country has set a target to increase its electricity generation capacity to 556 MW by 2024 [14]. Figure 1 shows the summary of electricity generation capacity by source as of June 2023. The shared capacity refers to the share of Rwanda from the electric power plants that it shares with its neighboring countries [14].

2.1.2 | Transmission and Distribution Network

The transmission system comprises high voltage (HV) lines with voltage levels of 220, 110, and 70 kV, as well as medium voltage (MV) lines with voltage levels of 30, 15, 17.32, and 5.5 kV. As of the end of June 2023, the total length of high-voltage transmission lines was approximately 1158 km. The 220 kV transmission lines were utilized for Rwanda's electricity network interconnection with Uganda in the north, Burundi in the south, and RDC in the west. Additionally, 33 operational substations nationwide exist for HV to MV power transformation. The electricity is distributed at both MV and LV. The distribution network is comprised of 10 520 km of MV and 18 465.7 km of LV distribution lines as shown in Table 1 [14, 15].

2.1.3 | Solar Energy Resources and Potentials

Identifying energy resources is a crucial first step in providing energy supply services. The availability of energy resources and their corresponding potential in a particular country plays a

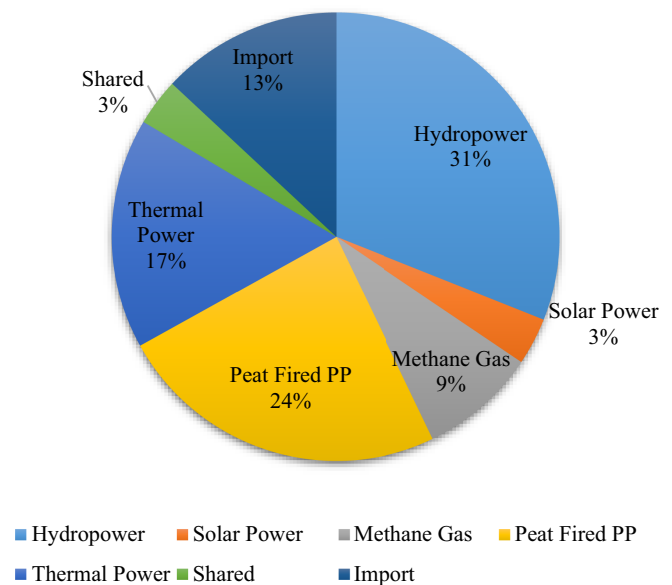


FIGURE 1 | Rwanda's power generation mix.

TABLE 1 | Transmission and distribution line voltage levels & solar energy resources and potentials.

		Length
Transmission and distribution line voltage levels		
High voltage	220 kV	1158 km
	110 kV	
	70 kV	
Medium voltage	30 kV	10 520 km
	17.32 kV	
	15 kV	
	5.5 kV	
Low voltage	0.4 kV	18 465.7 km
Solar energy resources and potentials		
Solar irradiation intensity	4 to 5.4 kWh/m ²	
Sun peak hours	5 h	
Total annual solar energy potential	66.8 TWh	
Daily PV potential	3.7 to 4.29 kWh/kWp	
Number of installed PV power plant	4	
Total installed capacity of PV power plant	12.05 MW	

significant role in ensuring sustainable, affordable, and accessible energy services. In Rwanda, the primary natural energy resources include water, solar radiation, methane gas, geothermal, and peat reserves. Rwanda's geographical location allows for ample sunlight, with solar irradiation intensity ranging from 4 to 5.4 kWh/m² and approximately 5 peak sun hours per day, as illustrated in Figure 2. The total annual solar energy potential is estimated to be 66.8 TWh, with a daily practical PV output potential between 3.7 and 4.29 kWh/kWp. Presently, four grid-connected solar power plants are in operation with a total installed capacity of 12.05 MW, contributing to 3% of the electric power installed capacity [16]. Table 1 presents the voltage levels of transmission and distribution lines, as well as the solar energy resources and potentials in Rwanda.

2.2 | Electricity Grid Reliability Assessment

The reliability of a power system encompasses both system adequacy and system security. System adequacy ensures that the system can meet customer energy demands by having sufficient generation, transmission, and distribution infrastructure. The system's security involves its ability to handle disruptions and maintain a stable and uninterrupted power supply [17]. Evaluating a power system's reliability involves assessing its capability to meet peak demand, its flexibility in addressing variability and uncertainty in both demand and generation, and its ability to maintain steady frequency and voltage within an acceptable range [18]. The degree of reliability is determined by factors such as the total number of electric interruptions (including the number of affected customers and connected loads), interrupted

power (measured in KVA), duration of interruptions, and frequency of interruptions [19]. A crucial approach to enhancing system reliability is to minimize the impact of one or more of these key influencing factors.

Reliability assessment is conducted at various hierarchical levels, encompassing the operational components of a power system, as illustrated in Figure 3. Hierarchical level I evaluates the adequacy of generation facilities and is referred to as "generation capacity reliability evaluation." In Hierarchical level II, the evaluation focuses on the reliability of generation and transmission facilities, which are also referred to as "bulk system or composite system reliability evaluation." On the other hand, Hierarchical level III assesses the reliability of all operational components of a power system and evaluates the reliability at the individual load point level [17, 20].

2.2.1 | Reliability Indices

The reliability indices can be classified as either load point or system-level indices. When evaluating the reliability of a power system, it is necessary to determine the frequency and duration of load interruptions during operation. Load point refers to a bus bar, terminal, or individual load and the corresponding indices provide insight into the reliability of each load point within the power system, helping to identify potential weak points. In contrast, system indices consolidate all load point indices. Moreover, reliability indices are further categorized as frequency-based or expectation-based, energy-related, and interruption cost-related indices. The System Reliability Indices include:

1. System Average Interruption Frequency Index (SAIFI), which measures the frequency of power interruptions experienced by customers over a specific period.
2. Customer Average Interruption Frequency Index (CAIFI), indicating the average frequency of interruptions experienced by affected customers.
3. Average System Interruption Frequency Index (ASIFI), which is based on load and reveals the distribution performance for systems serving industrial or commercial customers.
4. System Average Interruption Duration Index (SAIDI), showing the total duration of interruptions experienced by customers during a specified period.
5. Average Interruption Duration Index (CAIDI), calculating the average time required to reinstate service for customers.
6. Average System Interruption Duration Index (ASIDI), indicating the total duration of interruptions based on load during a specified period.
7. Average Service Availability Index (ASAI), representing the fraction of time that customers remain connected during a specified period.
8. Average Service Unavailability Index (ASUI), indicating the probability of all loads being supplied.

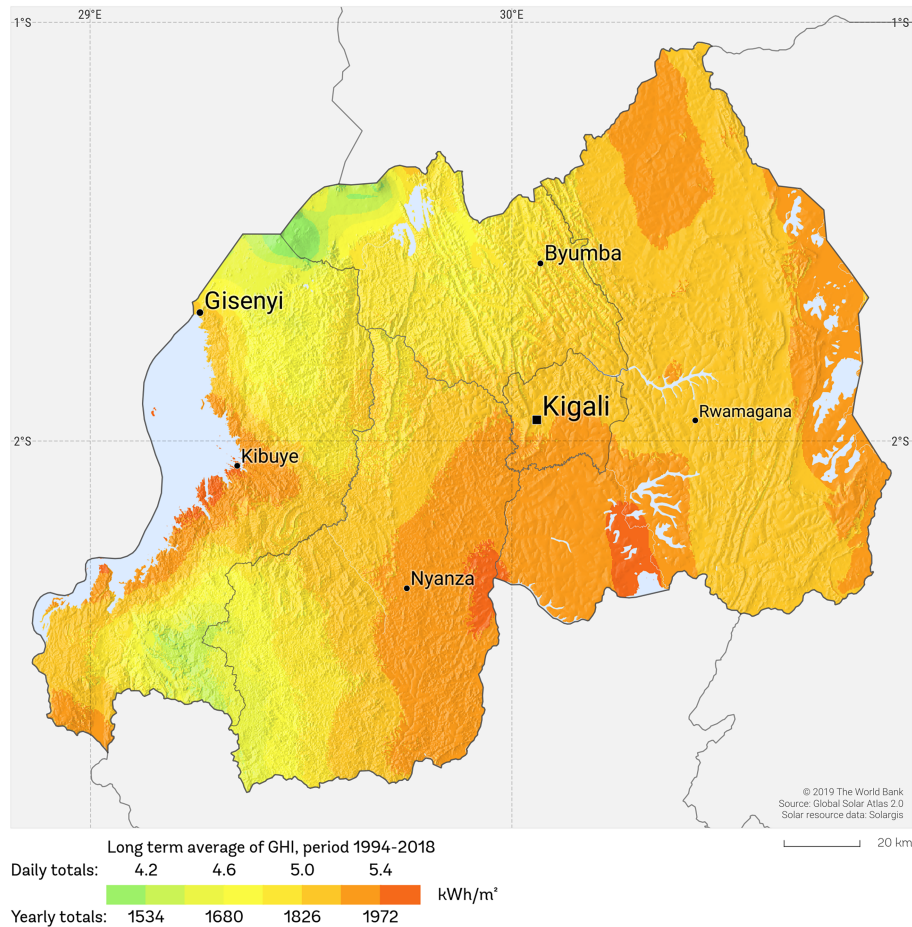


FIGURE 2 | Rwanda Global Solar Irradiation (<https://globalsolaratlas.info/download/rwanda>).

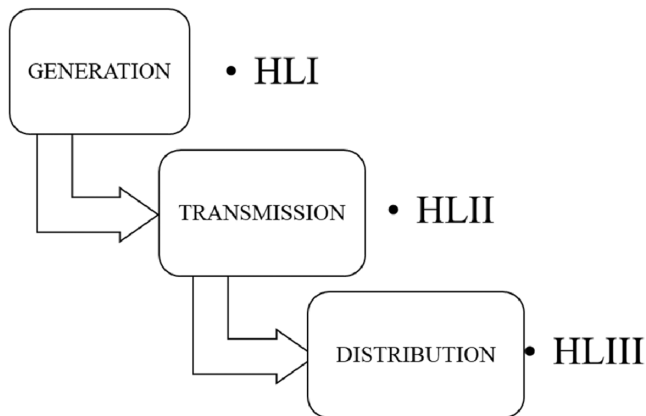


FIGURE 3 | Power system and hierarchical levels for reliability assessment.

Two methods are used to evaluate reliability: the historical and predictive methods. The historical method relies on the system's historical failure data. On the other hand, the predictive method involves analytical and simulation analyses. The analytical analysis uses a simplified mathematical model to calculate reliability indices. In contrast, simulation analysis utilizes the random behavior of the system to simulate a series of real experiments to estimate reliability indices. The Monte Carlo Simulation Method

(MCS) is a type of predictive method, also called multiple probability simulation, that generates random failures for each power system unit [21]. The MCS method is particularly well-suited for large and complex systems with high levels of uncertainty [21, 22]. Figure 4 illustrates the reliability analysis flow chart for randomly generated N_s samples and X stochastic variables with their respective marginal probability density functions for each realization i , element j , and time step t for stochastic variable $X^{i,j,t}$ [23].

2.3 | Review of Related Studies

Exploring the assessment of power system reliability has been a focal point in literature, with numerous research studies employing diverse methods and algorithms. These studies frequently utilized specialized programs or software to facilitate their analyses.

Wadi [19] evaluated the dependability of a closed-loop distribution system by employing an analytical approach that relied on Total Loss of Continuity (TLOC) and the Modified Monte Carlo Simulation (MMCS) method. The study indicates that over 80% of operational failures in a power system occur within the distribution network. As a result, there is a significant focus on assessing and enhancing the reliability of the distribution network. The study compared the effectiveness of two new techniques for assessing the reliability of a distribution system in Istanbul,

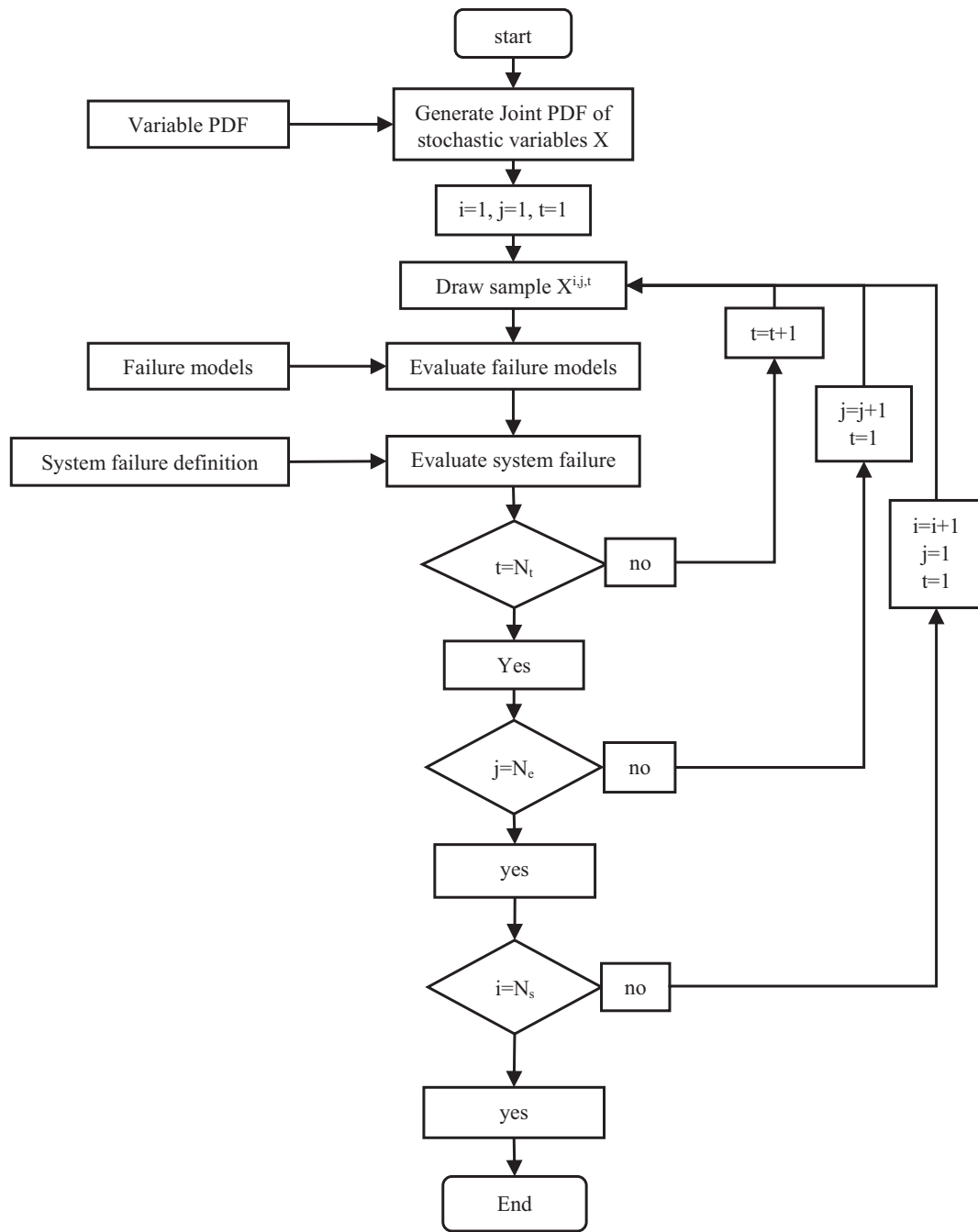


FIGURE 4 | Flow chart for reliability analysis based on Monte Carlo simulation method.

Turkey. These techniques were tested using actual data provided by the Bosphorus Electric Distribution Company and were compared with the results obtained through the Monte Carlo Simulation (MCS) method. The comparison revealed the practicality and adaptability of the two new techniques based on their results.

Saeed [24] examined how distributed generations (DG) affect the reliability of distribution systems. Utility companies are encouraged to improve both non-renewable and renewable distributed generations (DGs) to satisfy customers' increasing demands. The study uses the Monte Carlo simulation algorithm to evaluate the reliability of a distribution network incorporating distributed generators with battery storage systems. The reliability indices

used for evaluation were SAIFI, SAIDI, and Expected Energy Not Supplied (EENS). The simulation consisted of two distinct phases. The first phase entailed an assessment of the reliability of the distribution network in the absence of DGs. The second phase involved an evaluation of network reliability by considering the integration of DGs. Thus, reliability improvement was observed by inserting DGs into the network.

Gautam [17] conducted reliability studies on a distribution system integrated with energy storage. As distributed generation becomes more integrated into distribution systems, energy storage systems offer promising solutions for improving system reliability and utilizing renewable energy. This study introduces

a novel reliability model for energy storage systems using the Monte Carlo simulation algorithm and presents a new approach to integrating this developed model with distributed generation and time-dependent loads.

Heidari [20] examined the dependability of the power distribution system when DG units are present. The presence of DGs can enhance network reliability by decreasing the frequency and length of outages, although this is contingent on the operating mode. Therefore, the islanding mode of operation of DGs presents a solution to reduce the frequency of interruption of selected loads. This work introduces a novel sub-islanding approach that relies on two algorithms to pinpoint the precise location of DG units and identify fault locations. It also features a mixed-integer linear programming model for the optimal placement of sectionalizing switches and a new mixed-integer nonlinear programming model for determining the best number and location of protective and sectionalizing switches. Lastly, the work employs a contingency-based analytical technique for reliability assessment.

Teshome [25] presented a study on DG in improving power system reliability at Addis Centre Substation (ACS). The main causes of reliability problems at the ACS substation and a solution for improving them were investigated and proposed. In this study, Power Factory simulations for reliability assessment, before and after integrating DGs in the system, were conducted. The proposed solutions showed the capability of improving reliability at the ACS substation.

Koyi [26] studied the improvement of Nigeria's electricity grid reliability with renewable microgrids (MG). In the study, Nigeria's grid was modeled in Power Factory, and different scenarios to increase electricity availability were studied by connecting DGs and ESS to the appropriate position to meet the load demand.

The traditional single power flow has been changed into a bidirectional flow by incorporating renewable energy sources like wind and solar PV systems into a power system as distributed generators. Moreover, these sources improve the reliability of power systems by reducing outages in radial systems through increased generation capacity and decreased frequency and duration of interruptions. Numerous studies have highlighted the significant role of distributed generators in enhancing system reliability. Adefarati and Bansal [27] conducted a study to assess a microgrid power system's reliability and financial performance that integrates renewable energy sources (RES). In the study, wind turbine generators (WTG), PV, and BESS were used as DGs. Steele et al. [28] studied the impact of variable energy resources on power system reliability. Dadfar et al. [29] studied the reliability evaluation of a 20 kW solar power station in grid-connected mode. Wei et al. [30] studied the reliability evaluation of MG with PV and WTG hybrid systems. Jayesh and Sreeraj [31] studied the coordination of DG and demand response, using Modified Elephant Herd Optimization to address cost, environment, and reliability challenges for optimal MG operation.

Numerous methods for assessing and evaluating reliability have been developed and utilized in various research studies to serve specific purposes. Such methods are Markov model [17, 25, 27, 32–35], Monte Carlo simulation [24, 25, 36–40], modified Monte

Carlo [19, 41], continuous-time Markov chain [29], reliable block diagram [42], variance reduction [36], and new mixed-integer nonlinear programming model [20].

3 | Methodology

This study involves conducting reliability assessments and load flow computations for the electricity grid in Rwanda at the distribution level on the Gatumba and Ntongwe feeders by integrating PV system and BESS. The data for this research was obtained from the Rwanda Energy Group. The power characteristics of the PV system and BESS for each specific load were established by utilizing the PV*SOL simulation tool after analyzing the data and estimating the load profile. In PV*SOL simulations, three scenarios were considered: 50%, 100%, and 150% as the PV to consumption ratio. The BESS capacity was chosen to match the nighttime load demand, based on PV*SOL's recommendation. The method for simulating in PV*SOL is detailed in [43]. The modeling of Gatumba and Ntongwe feeders is done using DigSILENT PowerFactory. Load profiles, PV system, and BESS power profiles are added to the DigSILENT PowerFactory Model for all scenarios. Subsequently, quasi-dynamic and reliability simulations are conducted, with 6 cases considered, as detailed in Table 2.

The simulation steps include:

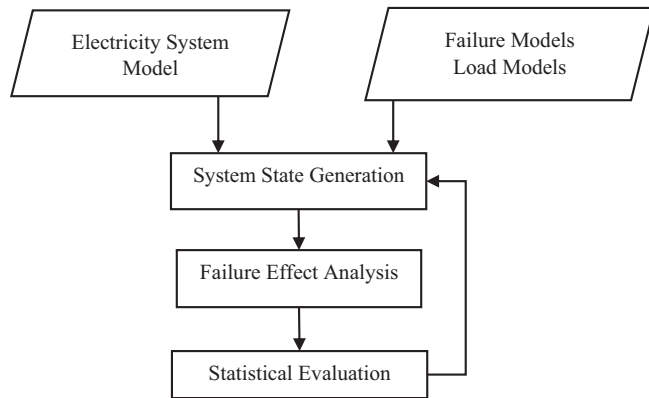
1. Estimating load profiles for Ntongwe and Gatumba Feeder's load points.
2. Performing PV*SOL simulations to size, analyze, and acquire the power characteristics of PV systems and BESS according to the scenarios specified in Table 2.
3. Modeling Ntongwe and Gatumba feeders in DigSILENT PowerFactory.
4. Importing load profiles, PV systems, and BESS power profiles into DigSILENT PowerFactory based on the cases in Table 2.
5. Performing quasi-dynamic and reliability analysis simulations in DigSILENT PowerFactory.

3.1 | NTONGWE and GATUMBA Feeders Configuration for Reliability Analysis

Compulsory and optional steps are configured in DigSILENT PowerFactory to complete the reliability assessment. The first step is to create failure stochastic models for lines, transformers, busbars, and switches. The second step is to define radial feeders. The third step is configuring switches for actuation time, switching stages, and power restoration. Finally, the number of connected customers, interruption cost, priority, and shedding load transfer are defined on the load side. In addition, optional steps are defining load characteristics, creating protection devices, creating maintenance plans, and specifying constraints. After configuration, the generation of reliability indices, with the help of the reliability analysis tool, consists of failure modeling, load modeling, system state creation, failure effect analysis (FEA), and statistical analysis, as shown in Figure 5. The failure models for

TABLE 2 | Considered cases for reliability analysis.

Scenario	Case	PV*SOL ratio to consumption	DigSILENT PowerFactory simulations description
SCENARIO-1	BASE CASE	50%	PV system
	CASE-1		PV system and BESS
SCENARIO-2	CASE-2	100%	PV system
	CASE-3		PV system and BESS
SCENARIO-3	CASE-4	150%	PV system
	CASE-5		PV system and BESS
	CASE-6		

**FIGURE 5** | Basic flow diagram for reliability analysis in DigSILENT PowerFactory.**TABLE 3** | Failure data for lines and transformers.

Element	Failure frequency in [1/(a*km)]	Repair duration in h
Line failure model	0.065	5
110 kV/30 kV-transformer failures model	0.015	15
30 kV/0.4 kV-transformer failures model	0.015	200

lines and transformers are configured according to the standard RBTS Bus-2 system, as the exact model data are unavailable, as shown in Table 3 [44]. The PV system and BESS's availability are calculated according to the power profile obtained from the PV*SOL simulation.

3.2 | Data Collection and Processing

3.2.1 | Data Collection

A comprehensive set of data and information about the grid is essential for conducting a reliability analysis of an electricity grid. Those data are input into simulation software that performs the reliability calculations. The necessary collected data include a single-line diagram of the national electricity network at Kigoma

Substation for Gatumba and Ntongwe Feeders, the geographical locations of secondary substations, the rated power of the transformers connected at Gatumba and Ntongwe Feeders, and the length of the distribution lines.

3.2.2 | Data Processing

3.2.2.1 | Load Point and Load Profile Estimation. In total, Gatumba and Ntongwe Feeders consist of 187 and 118 s distribution transformers, respectively. Distribution transformers in the same geographical location or on the same feeder branch are grouped together. The combined capacity of the grouped transformers is considered as the maximum load demand, and their number is considered as the number of connected customers. The load point location refers to the location of the distribution transformer with the highest capacity or the midpoint of all grouped distribution transformers. Figure 6 shows the geographical layout of Ntongwe and Gatumba feeders after the distribution transformers are grouped, along with a detailed connection of one PV system and BESS.

The types of loads connected at Ntongwe and Gatumba feeders are residential, commercial, industrial, and small industrial. The load profile of each load point is estimated by taking into account the maximum load demand and load type and using standard load profiles (SLP) from the Federal Association of the Germany Energy and Water Industries (BDEW). Equation (1) provides the estimated load profile (SLP):

$$ELP = \sum_{n=1}^P (MLD_n * SLP_n) \quad (1)$$

where ELP is the Estimated Load Profile, MLD is the Maximum Load Demand, n is the Load Type, P is the Number of Load Type, and SLP is the BDEW Standard Load Profile.

3.2.2.2 | Solar PV System and BESS Data. The Gatumba and Ntongwe feeders are integrated solar PV systems and BESS at selected load points. The integration of the PV system has been analyzed in three different scenarios: 50%, 100%, and 150% of the desired ratio to consumption. Each scenario has been thoroughly examined in two cases: one with just the PV system, and the other with the PV system in combination with BESS. Table 4 shows the list of connected PV systems and BESS capacity and associated loads for each case.

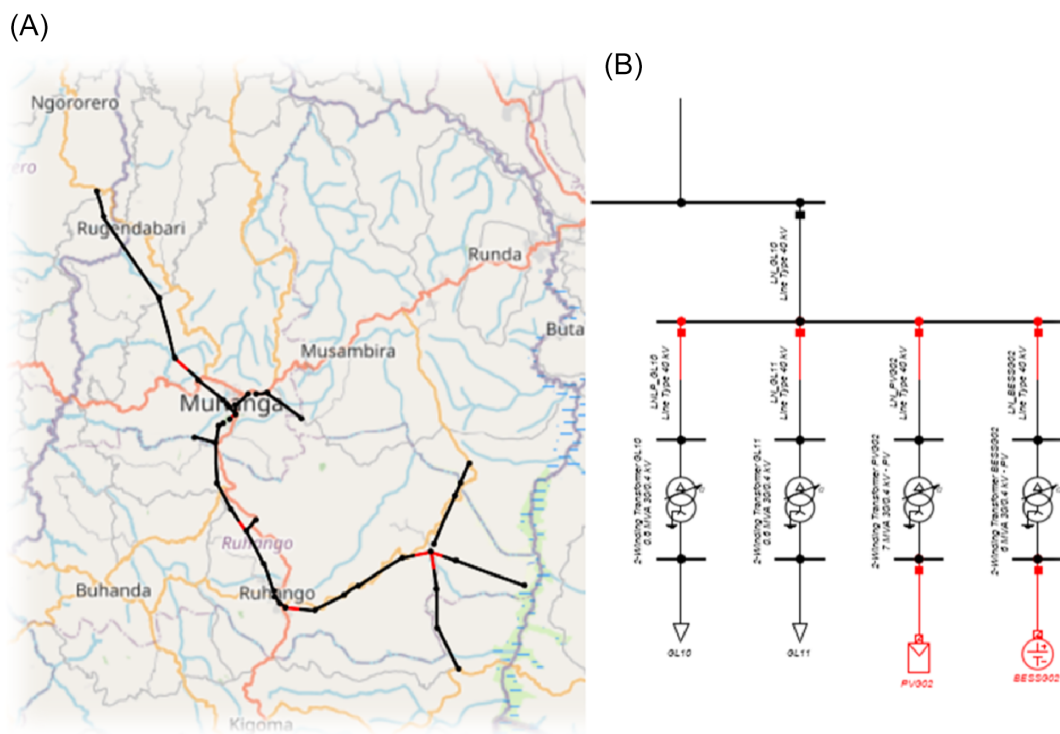


FIGURE 6 | (A) Geographical Representation of Ntongwe and Gatumba Feeders, (B) Detail Connection of PV system, BESS, and Load Connection at Distribution.

TABLE 4 | PV system and BESS capacity at each load center.

PV name	BESS name	Load name	PV peak power (MWp) for each case			BESS system		
			50%	100%	150%	No of battery systems	Battery inverter nominal output (kW)	Battery capacity (kWh)
PV-G01	BESS-G01	G-L01, G-L02, G-L03, G-L04, G-L05, G-L06, G-L07	4.81	9.62	14.43	14	300	1133.8
PV-G02	BESS-G02	G-L08, G-L09, G-L10 G-L11, G-L12, G-L13, G-L14	6.50	13.01	19.52	19	300	1133.8
PV-G03	BESS-G03	G-L15, G-L16, G-L17 G-L19, G-L19	3.27	6.54	9.82	10	300	1133.8
PV-G04	BESS-G04	G-L20, G-L21, G-L22 G-L23, G-L24, G-L25, G-L26	5.08	10.14	15.21	14	300	1133.8
PV-G05	BESS-G05	G-L27, G-L28, G-L29 G-L30, G-L31, G-L32, G-L33	4.35	8.70	13.05	13	300	1133.8
PV-N01	BESS-N01	N-L01, N-L02, N-L03, N-L04	1.93	3.86	5.80	6	300	1133.8
PV-N02	BESS-N02	N-L05, N-L06, N-L07, N-L08, N-L09, N-L10	3.18	6.36	9.54	10	300	1133.8
PV-N03	BESS-N03	N-L11, N-L12, N-L13, N-L14	1.63	3.26	4.89	5	300	1133.8
PV-N04	BESS-N04	N-L15, N-L16, N-L17	1.56	3.12	4.68	5	300	1133.8

4 | Simulation Results and Discussion

4.1 | PV*Sol Simulation

In the PV*SOL simulation tool, a simulation of a grid-connected PV system with electrical appliances and battery systems was conducted to evaluate the performance of a solar PV system and BESS. The simulation results summary is presented in Figures 7 and 8. Across all scenarios, the performance ratio exceeds 80%, demonstrating the high potential of solar energy in the installed locations. The desired ratio to consumption should be selected based on the level of self-sufficiency and own power consumption. For a 50% desired ratio to consumption, the level of self-sufficiency falls below 50%, indicating that the system is designed to meet only 50% of the load demand despite consuming

100% of its power. On the other hand, for 100% and 150% design ratios to consumption, the self-sufficiencies are around 72% and 82%, respectively, while the own power consumptions are approximately 82% and 60%, respectively.

4.2 | Quasi-Dynamic Simulation

During the first week of the year, quasi-dynamic simulations and load-flow calculations are conducted to analyze the generation profiles of PV systems and the power flow from the external grid, and it involves examining the behavior of the PV systems' energy generation and studying how the power is distributed and flows within the grid. Figure 9 shows the power flow from the external grid to Ntongwe and Gatumba feeders integrated with the PV System and BESS for all cases.

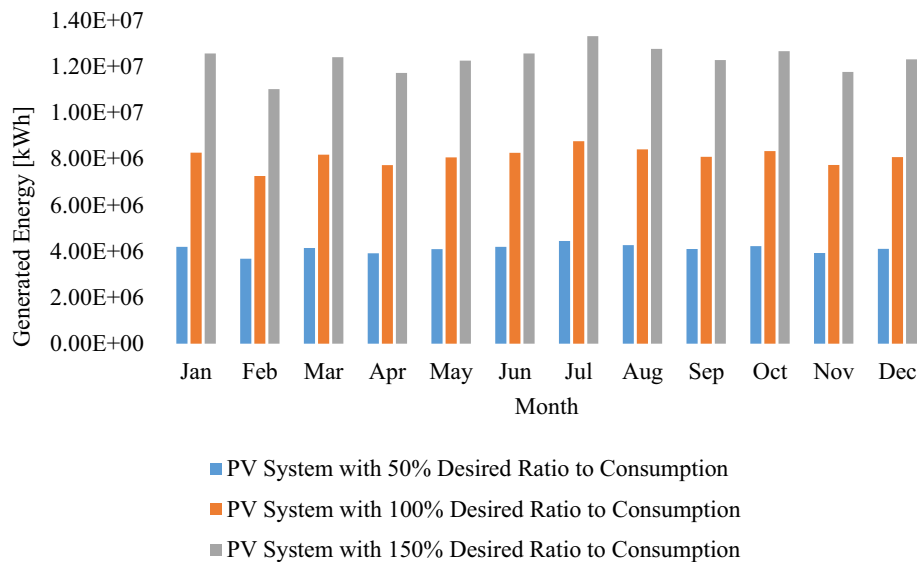


FIGURE 7 | PV system monthly total energy generation.

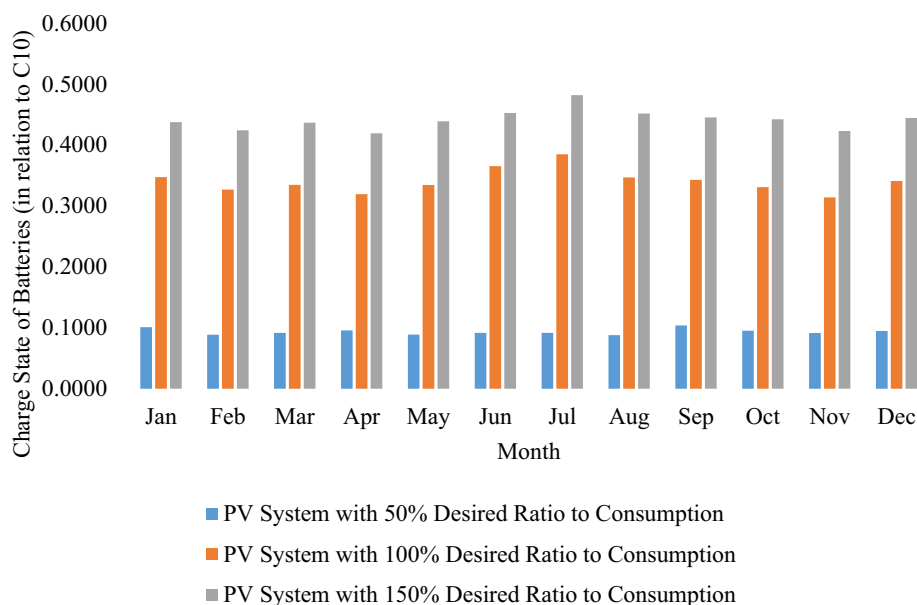


FIGURE 8 | Average monthly charge state of BESS (in relation to C10).

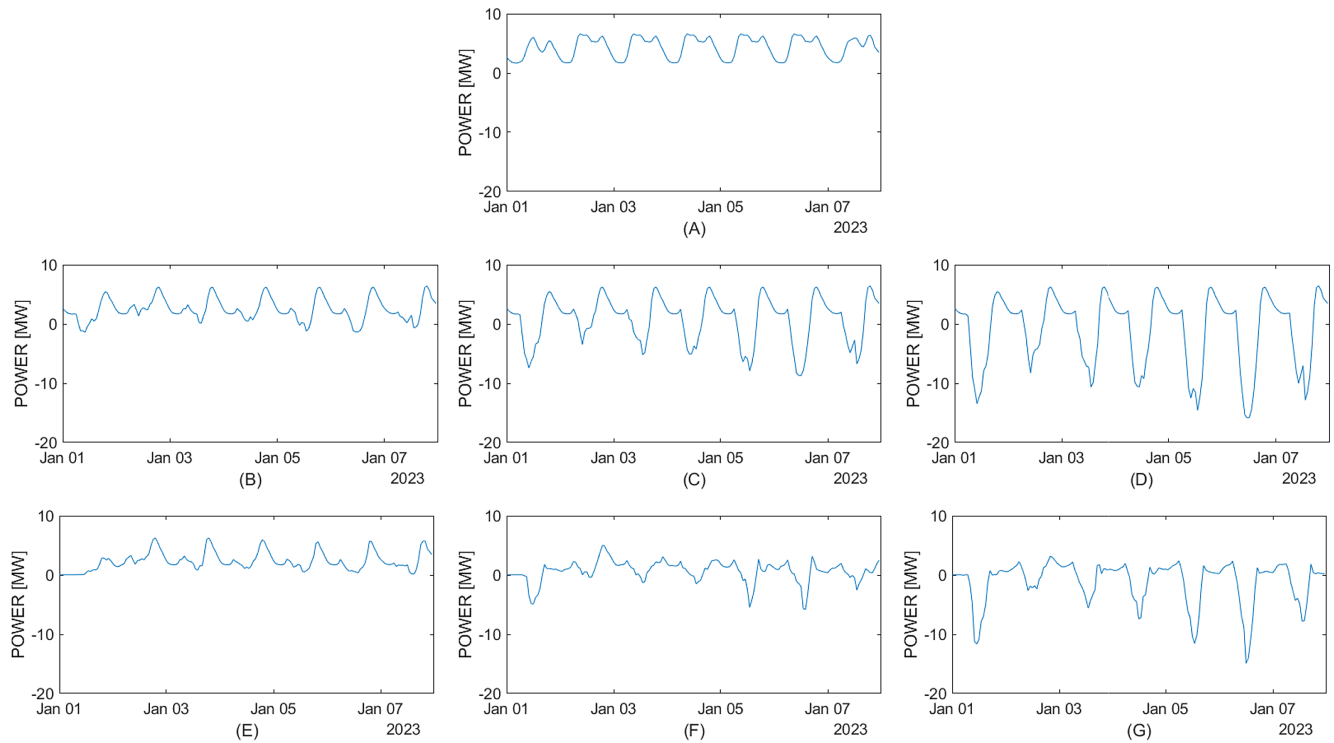


FIGURE 9 | Power supply from external grid to Ntongwe and Gatumba feeders: (A) Without PV system, (B) PV system with desired consumption ratio of 50%, (C) PV system with desired consumption ratio of 100%, (D) PV system with desired consumption ratio of 150%, (E) PV system and BESS with desired consumption ratio of 50%, (F) PV system and BESS with desired consumption ratio of 100%, (G) PV system and BESS with desired consumption ratio of 150%.

TABLE 5 | Comparison of reliability indices from simulation results.

Reliability	Base case	50% ratio to consumption		100% ratio to consumption		150% ratio to consumption	
		PV only	PV and BESS	PV only	PV and BESS	PV only	PV and BESS
SAIFI [1/Ca]	3.494631	0.990686	0.990686	0.990686	0.990686	0.990686	0.990686
CAIFI [1/Ca]	3.494631	0.990686	0.990686	0.990686	0.990686	0.990686	0.990686
SAIDI [h/Ca]	12.699	0.888	0.565	0.884	0.646	0.888	0.664
CAIDI [h]	3.634	0.897	0.57	0.892	0.652	0.897	0.67
ASAI	0.998550	0.999898	0.999935	0.999899	0.999926	0.999898	0.999924
ASUI	0.001449	0.000101	0.000064	0.000101	0.000073	0.000101	0.000075
ENS [MWh/a]	31.891	2.253	1.498	2.243	1.687	2.253	1.729
AENS [MWh/Ca]	0.105	0.007	0.005	0.007	0.006	0.007	0.006
ACCI [MWh/Ca]	1.676	0.035	0.023	0.035	0.026	0.035	0.027
ASIFI [1/a]	3.522847	0.887972	0.887972	0.887972	0.887972	0.887972	0.887972
ASIDI [h/a]	12.17744	0.860373	0.572025	0.856508	0.644141	0.860165	0.660365

4.3 | Reliability Analysis Results

The reliability analysis considers three different scenarios. Two cases are examined for each scenario: one with only a PV system and the other with a PV system and a BESS. The reliability analysis results for reliability indices are compared in Table 5. The indices are SAIFI, CAIFI, SAIDI, CAIDI, ASAI, ASUI, ENS, AENS, ACCI, ASIFI, and ASIDI.

The simulation results from Table 5 show that integrating PV systems or PV systems with BESS leads to improved reliability at Gatumba and Ntongwe feeders. Integration with a PV system and BESS results in a 71.6% improvement in both SAIFI and CAIFI. The reliability improves by 94% for SAIDI with the integration of the PV system alone and by 95.5% when BESS is added. For CAIDI, the reliability improves by 75% with integrating a PV system alone and reaches 84.3% by adding BESS. When only the PV

system is integrated, the reliability improvements are 92.9% for ENS, 93.3% for AENS, 74.7% for ASIFI, and 92.9% for ASIDI. Integrating the PV system and BESS results in a 95.3% improvement for ENS, 95.2% for AENS, 74.7% for ASIFI, and 95.3% for ASIDI.

5 | Conclusion

This study aims to evaluate the impact of integrating Solar PV and BESS at the distribution level to improve the reliability of the Gatumba and Ntongwe feeders. According to the simulation results from PV*SOL, the solar energy performance ratio is 80%, indicating a high potential for solar energy in the installed locations. The level of self-sufficiency varies based on the desired consumption ratio and can reach up to 82%. For a PV system with battery energy storage, the system needs to produce enough power to meet the load and also recharge the BESS by enhancing the targeted consumption ratio. Based on the simulation results from PV*SOL, the annual average charge state of the BESS in relation to C10 was 0.0935 with a 50% desired consumption ratio, 0.3405 with a 100% desired consumption ratio, and 0.4412 with a 150% desired consumption ratio. The analysis results from DigSILENT PowerFactory demonstrate that integrating a PV system with a BESS to the electric grid can significantly improve system reliability. The simulations show a 71.6% improvement in average interruption frequency for both customers and the system, a 95.5% improvement in system average interruption duration, and an 84.3% improvement in customer average interruption duration.

Author Contributions

Obed Nkuriyigoma: conceptualization, funding acquisition, writing – original draft, methodology, validation, writing – review and editing. **Engin Özdemir:** conceptualization, funding acquisition, writing – original draft, methodology, validation, supervision.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

1. L. Ahmad, N. Khordehghah, J. Malinauskaitė, and H. Jouhara, “Recent Advances and Applications of Solar Photovoltaics and Thermal Technologies,” *Energy* 207 (2020): 118254, <https://doi.org/10.1016/j.energy.2020.118254>.
2. E. Rakhshani, K. Rouzbehi, A. J. Sánchez, A. C. Tobar, and E. Pouresmaeil, “Integration of Large Scale PV-Based Generation Into Power Systems: A Survey,” *Energies* 12, no. 8 (2019): 1425, <https://doi.org/10.3390/en12081425>.
3. D. Engin and M. Çolak, “Modeling and Performance Optimization of Photovoltaic and Thermal Collector Hybrid System,” *Turkish Journal of Electrical Engineering and Computer Sciences* 24, no. 5 (2016): 3524–3542, <https://doi.org/10.3906/elk-1404-425>.
4. M. I. Chergui, “Improved Control of a Photovoltaic Pumping System by DTC-SVM to Optimize the Water Flow,” *Turkish Journal of*

Electrical Engineering and Computer Sciences 24, no. 3 (2016): 1474–1486, <https://doi.org/10.3906/elk-1311-151>.

5. G. C. Abdin and M. Noussan, “Electricity Storage Compared to Net Metering in Residential PV Applications,” *Journal of Cleaner Production* 176 (2018): 175–186, <https://doi.org/10.1016/j.jclepro.2017.12.132>.
6. A. Jasuan, Z. Nawawi, and H. Samaulah, “Comparative Analysis of Applications Off-Grid PV System and On-Grid PV System for Households in Indonesia,” in *2018 International Conference on Electrical Engineering and Computer Science (ICECOS)* (IEEE, 2018), 253–258, <https://doi.org/10.1109/ICECOS.2018.8605263>.
7. Y. Jiang, L. Kang, and Y. Liu, “Multi-Objective Design Optimization of a Multi-Type Battery Energy Storage in Photovoltaic Systems,” *Journal of Energy Storage* 39 (2021): 102604, <https://doi.org/10.1016/j.est.2021.102604>.
8. G. Stephens, C. Dieterle, E. Hossain, and R. Bayindir, “Feasibility Study of Grid-Connected Solar Plant: An In-Depth Analysis of System Modeling and Proper Technology Selection,” *International Journal of Electrical Engineering Education* 60, no. 2 (2023): 159–175, <https://doi.org/10.1177/0020720920928543>.
9. M. Shafiullah, S. D. Ahmed, and F. A. Al-Sulaiman, “Grid Integration Challenges and Solution Strategies for Solar PV Systems: A Review,” *IEEE Access* 10 (2022): 52233–52257, <https://doi.org/10.1109/ACCESS.2022.3174555>.
10. K. Boutaghane, N. Bennecib, M. Benidir, H. Benbouhenni, and I. Colak, “Performance Enhancement of a Three-Phase Grid-Connected PV Inverter System Using Fractional-Order Integral Sliding Mode Controls,” *Energy Reports* 11 (2024): 3976–3994, <https://doi.org/10.1016/j.egyr.2024.03.049>.
11. S. Ostovar, A. Esmaeili-Nezhad, M. Moeini-Aghtaie, and M. Fotuhi-Firuzabad, “Reliability Assessment of Distribution System With the Integration of Photovoltaic and Energy Storage Systems,” *Sustainable Energy, Grids and Networks* 28 (2021): 100554, <https://doi.org/10.1016/j.segan.2021.100554>.
12. A. T. Mohamed, A. A. Helal, and S. M. El Safty, “Distribution System Reliability Evaluation in Presence of DG,” *Sustainable Energy, Grids and Networks* 28 (2021): 100554, <https://doi.org/10.1109/EEEIC.2019.8783657>.
13. J. D. K. Hakizimana, S. P. Yoon, T. J. Kang, H. T. Kim, Y. S. Jeon, and Y. C. Choi, “Potential for Peat-To-Power Usage in Rwanda and Associated Implications,” *Energy Strategy Reviews* 13–14 (2016): 222–235, <https://doi.org/10.1016/j.esr.2016.04.001>.
14. Annual Report 2022–2023, Rwanda Energy Group, 2023, https://www.reg.rw/fileadmin/user_upload/REG_ANNUAL_REPORT_2022-2023.pdf.
15. “Distribution,” Rwanda Energy Group, <https://www.reg.rw/what-we-do/distribution/>.
16. E. Hakizimana, D. Sandoval, U. G. Wali, and K. Venant, “Inventory Analysis of Power Plants in Rwanda and Estimated Generation Capacities,” *IJSTR* 9, no. 12 (2020): 222–229.
17. P. Gautam, “Reliability Studies of Distribution Systems Integrated With Energy Storage”. Doctoral Dissertation, University of Saskatchewan, 2018.
18. U.S. Department of Energy, Maintaining Reliability in the Modern Power System, 2016, <https://www.energy.gov/sites/prod/files/2017/01/f34/Maintaining%20Reliability%20in%20the%20Modern%20Power%20System.pdf>.
19. M. J. Wadi, *Reliability Assessment of Closed Ring Power Distribution Systems* (Doctoral Dissertation, Yıldız Technical University, 2017).
20. A. Heidari, Reliability Analysis of Power Distribution System in Presence of Distributed Generation Units. Doctoral Dissertation, UNSW Sydney, 2015.

21. S. Garip, Ş. Özdemir, and N. Altın, "Power System Reliability Assessment - A Review on Analysis and Evaluation Methods," *Journal of Energy Systems* 6, no. 3 (2022): 401–419, <https://doi.org/10.30521/jes.1099618>.
22. M. S. Hashish, H. M. Hasanie, H. Ji, et al., "Monte Carlo Simulation and a Clustering Technique for Solving the Probabilistic Optimal Power Flow Problem for Hybrid Renewable Energy Systems," *Sustainability* 15, no. 1 (2023): 783, <https://doi.org/10.3390/su15010783>.
23. J. C. Pol, P. Kindermann, M. G. van der Krogt, et al., "The Effect of Interactions Between Failure Mechanisms on the Reliability of Flood Defenses," *Reliability Engineering and System Safety* 231 (2023): 108987, <https://doi.org/10.1016/j.res.2022.108987>.
24. S. Saeedi, "Impacts of Distributed Generations on Distribution System Reliability," Master's Thesis, Fen Bilimleri Enstitüsü, 2016.
25. A. D. Teshome, *Study of Distributed Generation in Improving Power System Reliability (Case Study: Addis Center Substation)* (School of Graduate Studies, Addis Ababa Institute of Technology, Addis Ababa University, 2016).
26. O. O. Koyi, "Improving the Reliability of Nigeria's Electricity Grid With Renewable Energy Microgrids," M. Sc. Thesis, 2019.
27. T. Adefarati and R. C. Bansal, "Reliability and Economic Assessment of a Microgrid Power System With the Integration of Renewable Energy Resources," *Applied Energy* 206 (2017): 911–933, <https://doi.org/10.1016/j.apenergy.2017.08.228>.
28. A. J. Harker Steele, J. W. Burnett, and J. C. Bergstrom, "The Impact of Variable Renewable Energy Resources on Power System Reliability," *Energy Policy* 151 (2021): 111947, <https://doi.org/10.1016/j.enpol.2020.111947>.
29. S. Dadfar, M. M. Marzban, and M. S. Ghafouri, "Reliability Evaluation of 20 kW Solar Power Station in Grid-Connected Mode (a Case Study)," in *2017 25th Iranian Conference on Electrical Engineering (ICEE)*, 2017, 1343–1348, <https://doi.org/10.1109/IranianCEE.2017.7985250>.
30. H. Wei, H. Zijun, T. H. Fengli, and Z. Li, "Reliability Evaluation of Microgrid With PV-WG Hybrid System," in *011 4th International Conference on Electric Utility Deregulation and Restructuring and Power Technologies (DRPT)* (IEEE, 2011), 1629–1632, <https://doi.org/10.1109/DRPT.2011.5994158>.
31. J. Priolkar and S. Es, "Distributed Generation and Demand Response Coordination for Optimal Operation of Microgrid Using Modified Elephant Herd Optimization," *Electric Power Components & Systems* (2024): 1–18, <https://doi.org/10.1080/15325008.2024.2330989>.
32. T. Adefarati and R. C. Bansal, "Reliability Assessment of Distribution System With the Integration of Renewable Distributed Generation," *Applied Energy* 185 (2017): 158–171, <https://doi.org/10.1016/j.apenergy.2016.10.087>.
33. L. Wu, C. Wen, and H. Ren, "Reliability Evaluation of the Solar Power System Based on the Markov Chain Method," *International Journal of Energy Research* 41, no. 15 (2017): 2509–2516, <https://doi.org/10.1002/er.3816>.
34. W. Li, L. Sun, W. Zou, and C. Luo, "Power System Reliability Analysis System Based on PSASP and Fault Enumeration Method and Applications," in *China International Conference on Electricity Distribution (CICED)*, 2014, 1323–1326, <https://doi.org/10.1109/CICED.2014.6991921>.
35. S. Garip, M. Bilgen, N. Altın, S. Özdemir, and I. Sefa, "Reliability Analysis of Microgrids: Evaluation of Centralized and Decentralized Control Approaches," *Electric Power Components & Systems* 51, no. 19 (2023): 2319–2338, <https://doi.org/10.1080/15325008.2023.2227174>.
36. A. Ali Kadhém, N. I. Abdul Wahab, I. Aris, J. Jasni, and A. N. Abdalla, "Computational Techniques for Assessing the Reliability and Sustainability of Electrical Power Systems: A Review," *Renewable and Sustainable Energy Reviews* 80 (2017): 1175–1186, <https://doi.org/10.1016/j.rser.2017.05.276>.
37. H. Wang, Z. Meng, Y. Guo, Y. Yao, and X. Xi, "Analysis of Micro-Grid Power Supply Reliability," in *China International Conference on Electricity Distribution (CICED)* (IEEE, 2014), 107–110, <https://doi.org/10.1109/CICED.2014.6991674>.
38. Y. Guo-hua, L. Yi, Y. Qi, and Y. Rong, "Study of Reliability of Grid Connected Photovoltaic Power Based on Monte Carlo Method," in *2011 IEEE Power Engineering and Automation Conference*, vol. 1 (IEEE, 2011), 92–95, <https://doi.org/10.1109/PEAM.2011.6134803>.
39. Z. Dongmei, S. Wenhao, and Z. Xu, "Reliability Evaluation for Mudanjiang Regional Power Grid," in *POWERCON 2014–2014 International Conference on Power System Technology: Towards Green, Efficient and Smart Power System, Proceedings* (IEEE, 2014), 746–751, <https://doi.org/10.1109/POWERCON.2014.6993858>.
40. N. R. Battu, A. R. Abhyankar, and N. Senroy, "Reliability Compliant Distribution System Planning Using Monte Carlo Simulation," *Electric Power Components & Systems* 47, no. 11–12 (2019): 985–997, <https://doi.org/10.1080/15325008.2019.1659454>.
41. M. Wadi, M. Baysal, A. Shobole, and M. R. Tur, "Reliability Evaluation in Smart Grids via Modified Monte Carlo Simulation Method," in *7th International IEEE Conference on Renewable Energy Research and Applications, ICRERA* (IEEE, 2018), 841–845, <https://doi.org/10.1109/ICRERA.2018.8566982>.
42. B. Bousshousa and A. Elmaouhab, "Smart Grid Reliability Using Reliable Block Diagram Case Study: Adrar's Isolated Network of Algeria," in *5th International Conference on Power Generation Systems and Renewable Energy Technologies, PGSRET* (IEEE, 2019), 26–27, <https://doi.org/10.1109/PGSRET.2019.8882711>.
43. O. Nkuriyigoma, E. Özdemir, and S. Sezen, "Techno-Economic Analysis of a PV System With a Battery Energy Storage System for Small Households: A Case Study in Rwanda," *Frontiers in Energy Research* 10 (2022): 957564, <https://doi.org/10.3389/fenrg.2022.957564>.
44. R. N. Allan, R. Billinton, I. Sjarief, L. Goel, and K. S. So, "A Reliability Test System for Educational Purposes: Basic Distribution System Data and Results," *IEEE Transactions on Power Systems* 6, no. 2 (1991): 813–820, <https://doi.org/10.1109/59.76730>.